
DSLA: DIRECTIONAL AND SIMILARITY-AWARE LATENT ALIGNMENT FOR PREFERENCE OPTIMIZATION

On the nature of what separates chosen from rejected

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ABSTRACT

Preference optimization learns from comparisons between chosen and rejected responses to a shared prompt. This supervision specifies the desired output ranking, while the internal features supporting that ranking are constrained indirectly through the output loss. The resulting ambiguity permits a model to reproduce the desired ordering through response length, formatting, or other features that correlate with preference labels without capturing the intended behavioral distinction. To make the underlying representational relationship explicit, we introduce Directional and Similarity-aware Latent Alignment (DSLA), a framework that supervises the hidden-state geometry of prompts, chosen responses, and rejected responses during preference training. DSLA couples a prompt–response similarity margin with a batch preference-direction term, encouraging chosen responses to have greater angular similarity to their prompts and encouraging agreement among chosen–rejected displacement directions within a batch. The resulting regularizer augments Direct Preference Optimization and Odds Ratio Preference Optimization through pooled hidden states, retaining the underlying output objective and its training structure. We analyze the bounded latent margins, approximate scale invariance, and conditions under which latent supervision remains active after the output preference margin saturates. We then evaluate the method through matched training comparisons, hidden-state diagnostics, generation analysis, and ablations of extraction depth, regularization strength, and pooling. Across the reported comparisons, DSLA increases chosen–rejected representation separation. The shortcut analyses further show reduced correlation with selected surface features in the evaluated settings, while the layer ablations favor later residual-stream representations. Generation diagnostics vary across backbones and datasets, establishing that stronger latent separation alone is insufficient to establish better response quality. These findings support explicit representation supervision as a practical mechanism for shaping preference geometry and motivate further investigation of how that geometry contributes to behavior.

Keywords Preference optimization · Representation learning · Language model alignment · Hidden-state geometry · Direct Preference Optimization · Odds Ratio Preference Optimization

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1 Introduction

Preference optimization teaches a policy to distinguish a chosen response from a rejected response. The distinction is expressed through an output objective, but the preference itself originates in a judgment about the response: its relevance to the prompt, the quality of its reasoning, its factual content, or the behavior it expresses. The training label records the outcome of that judgment without necessarily identifying the features that produced it. This leaves a central question unresolved: what internal representation does the model learn when it learns to prefer an answer?

Direct Preference Optimization (DPO) [1] and Odds Ratio Preference Optimization (ORPO) [2] provide practical formulations for learning from preference pairs. DPO optimizes a policy–reference log-probability margin, while ORPO combines the chosen–response language-modeling loss with an odds-ratio preference penalty. Both objectives update the model’s hidden states through backpropagation, yet neither explicitly specifies the geometry that should support the preference decision. Consequently, similar output margins can emerge from different internal organizations of the same chosen and rejected responses.

This representational freedom matters when preference labels correlate with features that are easier to learn than the intended criterion. A chosen response may be longer, use fewer lists, or adopt a more recognizable tone than its rejected counterpart. Such features can help predict the label within the training distribution without establishing that the model has learned the quality judgment behind it. Shortcut learning [3] describes this broader problem, and length exploitation in DPO [4] demonstrates its relevance to preference optimization. A successful output ranking therefore leaves open the mechanism by which the model arrived at that ranking.

We approach this problem by supervising the relationship among the prompt representation and the representations of its chosen and rejected responses. Work on activation steering and behavioral direction ablation shows that specific language-model behaviors can be influenced through directions in the residual stream [5, 6]. Our previous Gabliteration work [7] studies this perspective through adaptive projections onto behavioral subspaces. These findings motivate a training question: if hidden-state geometry participates in the expression of behavior, can we make that geometry an explicit target of preference learning?

To investigate this question, we introduce *Directional and Similarity-aware Latent Alignment* (DSLA). The similarity component encourages the chosen response to have greater cosine similarity to the prompt than the rejected response. The directional component encourages the chosen–rejected displacements within a batch to agree with their mean direction. Together, these components define an auxiliary latent objective that can be added to DPO or ORPO. The preference loss continues to supervise the output decision, while the latent loss places an additional constraint on the representations supporting that decision. Because the margins depend primarily on angles, increasing hidden-state norms alone does not satisfy the latent objective away from the numerical stabilizers.

The geometric formulation gives us a direct way to inspect the effect of the additional supervision. We compare DSLA with matched baselines across language-model families and preference datasets, then examine output losses, hidden-state separation, and generated responses. We also vary the extraction layer, latent weight, and pooling operator to determine which representations the regularizer acts on and how that choice affects training. In the reported comparisons, DSLA increases chosen–rejected separation even when ordinary preference-loss curves remain close. The shortcut analyses show lower correlation with selected surface features in the evaluated settings, and the extraction-layer study favors the late and final residual stream. These observations establish a change in preference geometry; interpreting that geometry as the intended semantic criterion requires additional behavioral evidence.

Generation provides a complementary view of the intervention. The diagnostics show gains, ties, and regressions depending on the backbone, objective, and dataset, so larger latent separation is not sufficient to infer better generations. We therefore study the geometric effect and its behavioral consequences together. The resulting contribution is an explicit, implementable formulation of latent preference supervision, with mathematical analysis and experiments that identify both its operation and its present limits.

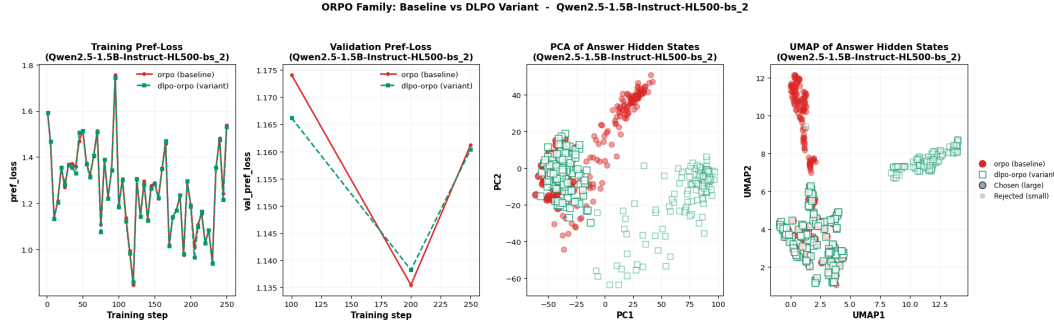


Figure 1: **Hidden-state geometry under latent preference supervision.** Matched ORPO and DSLA-ORPO runs have similar training and validation preference-loss curves, while their PCA and UMAP projections show different organizations of answer hidden states. The projections illustrate a representational effect that is less apparent in the scalar loss.

1.1 Contributions

Our contributions are:

1. **Explicit latent preference supervision.** A similarity and direction regularizer for DPO and ORPO that supervises hidden-state relationships among prompts, chosen responses, and rejected responses.
2. **Analysis of the latent objective.** Bounded-margin and approximate scale-invariance properties, latent gradient pressure after output-margin saturation, and directional estimation under explicit assumptions.
3. **Matched geometric and generation comparisons.** Increased chosen-rejected separation and reduced surface-feature correlation across the reported comparisons, alongside generation outcomes that vary with training settings.
4. **Extraction and regularization ablations.** Studies of layer selection, latent weight, and pooling, and find that these choices affect representation geometry even when the output objective offers little distinction among configurations.

2 Related Work

2.1 Preference Optimization and Output Supervision

Learning from human preferences provides supervision for behaviors that are difficult to specify through a token target alone. Early preference-based reinforcement learning [8] and subsequent language-model RLHF systems [9, 10] learn a reward from comparative feedback and optimize the policy against that reward. DPO [1] derives a direct training objective from the relationship between a KL-regularized optimal policy and its reward, using a Bradley-Terry preference model [11]. ORPO [2] incorporates an odds-ratio preference penalty into supervised fine-tuning and removes the separate reference policy.

Related formulations change how output supervision is expressed. SimPO [12] uses average sequence log-probability as a reference-free reward and introduces a target reward margin. KTO [13] uses a prospect-theoretic utility formulation and learns from binary desirability labels. These approaches differ in reference dependence, data requirements, and utility models, but supervise preference primarily through output probabilities. DSLA complements this signal with an objective defined over hidden representations. We instantiate it with DPO and ORPO, while compatibility with other preference objectives remains an empirical question.

2.2 Shortcut Learning and Preference Proxies

Geirhos et al. [3] describe shortcut learning as the use of decision rules that succeed under observed evaluation conditions but fail to transfer to more demanding settings. In preference learning, response length and stylistic cues can become such proxies when they correlate with labels. Park et al. [4] study length exploitation in DPO and develop a regularization strategy to disentangle length from quality. Their analysis highlights that improving a preference objective may also reinforce biases present in the feedback signal.

DSLA does not assume a predefined shortcut feature; consequently, its learned geometry may itself reflect unintended correlations in the preference data. We therefore examine surface-feature correlations and controlled response perturbations separately from downstream behavioral evaluation.

2.3 Activation Steering and Behavioral Subspaces

Activation engineering [5] controls generation through vector additions to intermediate activations. Representation engineering [14] develops a broader framework for reading and manipulating population-level representations associated with high-level behavior. Arditi et al. [6] identify a direction whose removal suppresses refusal and whose addition elicits it in the studied models. These interventions provide evidence that particular behaviors can depend on identifiable structure in the residual stream.

Our Gabliteration framework [7] extends behavioral weight modification through adaptive multidirectional extraction, regularized projections, and layer selection. Unlike these post-hoc interventions, DSLA estimates preference directions during training and uses them as an optimization signal rather than directly modifying activations or weights.

2.4 Learning Through Hidden Representations

Representation Finetuning (ReFT) [15] learns task-specific interventions on the hidden representations of a frozen policy model, with LoReFT using a low-rank subspace formulation. Whereas ReFT parameterizes an intervention in representation space, DSLA applies a relational objective to representations produced by the policy; in our experiments, the resulting gradients update LoRA adapters.

The Image-based Joint-Embedding Predictive Architecture (I-JEPA) [16] learns image representations by predicting target-block embeddings from context-block embeddings. It provides a broader precedent for treating representations themselves as explicit learning targets, although its predictive objective is fundamentally different from DSLA’s preference-conditioned geometry.

3 Background and Definitions

3.1 Notation and Formal Setting

We consider a preference dataset

$$\mathcal{D} = (x_i, y_i^+, y_i^-)_{i=1}^n, \quad (1)$$

where x_i is a prompt, y_i^+ is the chosen response, and y_i^- is the rejected response. A policy π_θ assigns conditional probabilities

$$\log \pi_\theta(y \mid x) = \sum_{t=1}^{|y|} \log \pi_\theta(y_t \mid x, y_{<t}), \quad (2)$$

and produces hidden states $H_\theta^{(\ell)}(x, y) \in \mathbb{R}^{T \times d}$ at layer ℓ , where T is the sequence length and d is the hidden dimension. We extract the final residual stream by default and study the extraction depth in Section 6.5. The layer determines which stage of the model’s computation receives direct latent supervision.

Definition 1 (Masked pooling operator). *Let $H \in \mathbb{R}^{T \times d}$ be a sequence of hidden states and let $m \in 0, 1^T$ be a token mask. We define the masked mean pooler*

$$\text{Pool}_m(H) = \frac{\sum_{t=1}^T m_t H_t}{\max\left(\sum_{t=1}^T m_t, 1\right)}. \quad (3)$$

In the default DSLA configuration, m is the response loss mask; the ablations additionally consider last-token pooling, last- k mean pooling, and prompt-plus-answer pooling.

Lemma 1 (Gradient through masked mean pooling). *Let $H \in \mathbb{R}^{T \times d}$ be hidden states and $m \in 0, 1^T$ a mask with $M = \sum_{t=1}^T m_t > 0$. For the pooled vector $c = \text{Pool}_m(H) \in \mathbb{R}^d$, the Jacobian with respect to each token’s hidden state is*

$$\frac{\partial c}{\partial H_t} = \frac{m_t}{M} I_d. \quad (4)$$

Consequently, the gradient of any loss $\mathcal{L}$ with respect to H_t is uniform over all active mask positions:

$$\nabla_{H_t} \mathcal{L} = \frac{m_t}{M} \nabla_c \mathcal{L}. \quad (5)$$

Proof. Differentiating Eq. 3 term-wise gives $\partial c / \partial H_t = m_t / M$ for each coordinate. The chain rule yields the gradient form immediately. $\square$

Interpretation. Masked mean pooling distributes the gradient with respect to the pooled vector equally across the active token positions. The mask therefore determines which hidden states receive the direct latent correction, while the active-span length determines its per-token magnitude. Answer-only pooling applies this supervision to the completion; including prompt positions also applies it to the prompt representations.

For each preference pair (x_i, y_i^+, y_i^-) we let $T_i^\pm$ denote the tokenized concatenation of the rendered chat-template prompt with the response $y_i^\pm$, and let $|T_i^\pm|$ be its length. Over the $T_i^\pm$ positions we define three binary masks: the *prompt-prefix mask* $m_x^\pm \in \{0, 1\}^{|T_i^\pm|}$, which is 1 on the full tokenized rendered-prompt prefix and 0 on the response suffix, and the *response masks* $m_+, m_- \in \{0, 1\}^{|T_i^\pm|}$ which are the complementary indicators restricted to the response suffix. With these masks in hand we define the three per-pair representations used by the similarity and direction terms. The two prompt-prefix pools are

$$p_i^{(+)} \triangleq \text{Pool}_{m_x^+}(H_\theta(x_i, y_i^+)), \quad p_i^{(-)} \triangleq \text{Pool}_{m_x^-}(H_\theta(x_i, y_i^-)), \quad (6)$$

and the two response pools are

$$c_i \triangleq \text{Pool}_{m_+}(H_\theta(x_i, y_i^+)), \quad r_i \triangleq \text{Pool}_{m_-}(H_\theta(x_i, y_i^-)). \quad (7)$$

The prompt representation p_i is the average of the two prompt-prefix pools from Eq. 6:

$$p_i = \frac{1}{2}(p_i^{(+)} + p_i^{(-)}), \quad (8)$$

which is in general *not* equal to $\frac{1}{2}(c_i + r_i)$ because $p_i^{(\pm)}$ is read with the prompt-prefix mask $m_x^\pm$ over the prompt positions of the concatenated sequence, whereas c_i and r_i are read with the response masks $m_\pm$ over the response suffix of the same sequence. The construction of $m_x^\pm$ as the full tokenized rendered-prompt prefix is deterministic and uses the same tokenization step that produces the response loss masks; the symmetric estimator therefore does not require an additional prompt-only forward pass.

Assumption 1 (Prompt stability). *For any preference pair (x_i, y_i^+, y_i^-) , the prompt representations obtained from the chosen and rejected sequences satisfy*

$$\left\| \text{Pool}_{m_x^+}(H_\theta(x_i, y_i^+)) - \text{Pool}_{m_x^-}(H_\theta(x_i, y_i^-)) \right\|_2 \leq \delta, \quad (9)$$

for some $\delta > 0$ independent of the response suffix.

Remark. Because cosine similarity is 1-Lipschitz with respect to unit-vector perturbations in its first argument, a deviation of δ in the prompt representation changes $s_i(\theta)$ by at most $O(\delta)$. The symmetric estimator p_i is therefore robust to small differences between the prompt contexts induced by the chosen and rejected sequences.

We use cosine similarity

$$\cos(a, b) = \left\langle \frac{a}{\sqrt{\|a\|_2^2 + \varepsilon}}, \frac{b}{\sqrt{\|b\|_2^2 + \varepsilon}} \right\rangle, \quad (10)$$

with a small numerical $\varepsilon > 0$ (we use $\varepsilon = 10^{-6}$).

Definition 2 (Output-level preference margin). *For DPO with reference model π_{ref} , the output-level margin is*

$$\Delta_{\text{DPO},i}(\theta) = [\log \pi_\theta(y_i^+ | x_i) - \log \pi_\theta(y_i^- | x_i)] - [\log \pi_{\text{ref}}(y_i^+ | x_i) - \log \pi_{\text{ref}}(y_i^- | x_i)]. \quad (11)$$

For ORPO, the corresponding margin is an odds-ratio margin computed from per-token mean log-probabilities. Let $\bar{\ell}_{\theta,i}^+$ and $\bar{\ell}_{\theta,i}^-$ denote the masked mean token log-probabilities for the chosen and rejected responses, and define the numerically stabilized logit

$$\text{logit}_\varepsilon(z) = \log \frac{\min\{z, 1 - \varepsilon\}}{1 - \min\{z, 1 - \varepsilon\}}. \quad (12)$$

The ORPO margin is then

$$\Delta_{\text{ORPO},i}(\theta) = \text{logit}_\varepsilon(\exp \bar{\ell}_{\theta,i}^+) - \text{logit}_\varepsilon(\exp \bar{\ell}_{\theta,i}^-), \quad (13)$$

where we use $\varepsilon = 10^{-6}$.

We use *output-level* for margins computed from policy probabilities and *representation-level* for the hidden-state margins defined in Section 4. This distinction identifies where each objective applies its supervision.

4 Directional and Similarity-aware Latent Alignment

We augment the preference objective with a latent regularizer that supervises the chosen–rejected relation in hidden-state space. The combined objective is

$$\mathcal{L}_{\text{DSLA}}(\theta) = \mathcal{L}_{\text{pref}}(\theta) + \lambda \mathcal{L}_{\text{latent}}(\theta), \quad \lambda \geq 0. \quad (14)$$

The preference term is either DPO,

$$\mathcal{L}_{\text{DPO}}(\theta) = \mathbb{E}_i [\text{softplus}(-\beta \Delta_{\text{DPO},i}(\theta))], \quad (15)$$

or ORPO,

$$\mathcal{L}_{\text{ORPO}}(\theta) = \mathbb{E}_i \left[-\bar{\ell}_{\theta,i}^+ \right] + \alpha \mathbb{E}_i [\text{softplus}(-\Delta_{\text{ORPO},i}(\theta))]. \quad (16)$$

We average the latent components when both are enabled:

$$\mathcal{L}_{\text{latent}} = \frac{1}{2} (\mathcal{L}_{\text{sim}} + \mathcal{L}_{\text{dir}}). \quad (17)$$

When a single component is enabled, $\mathcal{L}_{\text{latent}}$ is that component alone: $\mathcal{L}_{\text{sim}}$ in similarity-only mode or $\mathcal{L}_{\text{dir}}$ in direction-only mode. The coefficient λ controls the contribution of latent supervision relative to the output objective.

4.1 Prompt – Response Similarity Margin

The first component encourages the chosen representation to be more compatible with the prompt representation than the rejected representation:

$$s_i(\theta) = \cos(p_i, c_i) - \cos(p_i, r_i), \quad (18)$$

$$\mathcal{L}_{\text{sim}}(\theta) = \mathbb{E}_i [\text{softplus}(\gamma(m - s_i(\theta)))], \quad (19)$$

where $m > 0$ is the latent margin and $\gamma > 0$ controls the sharpness of the soft margin. This loss is a smooth relaxation of the constraint

$$\cos(p_i, c_i) \geq \cos(p_i, r_i) + m. \quad (20)$$

Interpreting the similarity margin. The margin in Eq. 18 measures an angular relationship among pooled hidden states. Its interpretation depends on what those states encode: response length, formatting, hedging, and lexical overlap can all contribute to the geometry. We combine the similarity term with the output preference objective and the directional term to supervise complementary relationships among the same preference pairs. The objective itself does not identify which features represent the intended criterion. We examine its association with measured surface features in Section 6.8.

4.2 Batch Preference-Direction Term

The second component estimates a batch-level preference direction. For a mini-batch B , define

$$d_i = c_i - r_i, \quad \bar{d}_B = \frac{1}{|B|} \sum_{i \in B} d_i, \quad u_B = \frac{\bar{d}_B}{\max(\|\bar{d}_B\|_2, 10^{-8})}. \quad (21)$$

DSLA then maximizes the alignment between each normalized pairwise direction and the batch direction:

$$a_i(\theta) = \left\langle \frac{d_i}{\sqrt{\|d_i\|_2^2 + \varepsilon}}, u_B \right\rangle, \quad (22)$$

$$\mathcal{L}_{\text{dir}}(\theta) = \mathbb{E}_{i \in B} [-\log \sigma(\gamma a_i(\theta))] = \mathbb{E}_{i \in B} [\text{softplus}(-\gamma a_i(\theta))]. \quad (23)$$

The batch mean makes directional agreement a relation among preference pairs. With multiple pairs, the loss encourages each displacement to align with their shared estimate. At batch size one, the estimate reduces to the example’s own direction and the directional component carries little discriminative information. Batch composition also matters: when a corpus combines distinct preference criteria, a shared mean can pull an example toward a direction that is poorly suited to its task. Task-specific estimates or a mixture of directions would relax this assumption; we leave their evaluation to future work.

Algorithm 1 DSLA training step

Require: Mini-batch $B = \{(x_i, y_i^+, y_i^-)\}$, policy π_θ , optional reference π_{ref} , latent weight λ , margin m , scale γ

Ensure: Stochastic gradient estimate for $\mathcal{L}_{\text{DSLA}}$

- 1: Run π_θ on (x_i, y_i^+) and (x_i, y_i^-) to obtain logits and hidden states.
 - 2: Compute masked sequence log-probabilities using Eq. 2.
 - 3: Compute $\mathcal{L}_{\text{pref}}$ using Eq. 15 for DPO or Eq. 16 for ORPO.
 - 4: Pool prompt, chosen, and rejected hidden states using Eq. 3.
 - 5: Compute $\mathcal{L}_{\text{sim}}$ from Eq. 19.
 - 6: Compute the batch preference direction u_B and $\mathcal{L}_{\text{dir}}$ from Eq. 23.
 - 7: Return $\nabla_\theta (\mathcal{L}_{\text{pref}} + \lambda(\mathcal{L}_{\text{sim}} + \mathcal{L}_{\text{dir}})/2)$.
-

Complexity. Let B be the batch size, T the sequence length, d the hidden dimension, and V the vocabulary size. The additional DSLA computation after the model forward pass is $O(BTd)$ for pooling and $O(Bd)$ for cosine and direction terms; the dominant cost remains the transformer forward and the log-softmax over vocabulary. DSLA-DPO additionally requires reference-model forward passes exactly as ordinary DPO does, whereas DSLA-ORPO remains reference-free.

Implementation. DSLA is officially supported in MLX-LM-LoRA [17] version 4.0.0 and later, and in the MLX-LoRA-Studio application version 3.0.0 and later. The implementation uses the masking, pooling, and prompt-prefix operations defined in Section 3.

5 Theoretical Analysis

We analyze how the latent objective constrains preference geometry. The results establish bounds on its margins, approximate scale invariance, and the gradient pressure that remains when an output margin saturates. We then study directional estimation and the relationship between latent displacement and output behavior under explicit assumptions. The analysis distinguishes properties of the stabilized objective from statements that depend on a shared preference direction existing in the representation.

Definition 3 (Shortcut policy). *A policy π_θ is a shortcut policy for a preference dataset $\mathcal{D}$ if there exists a superficial feature extractor $\psi(y) \in \mathbb{R}^k$ and a function $f : \mathbb{R}^k \rightarrow \mathbb{R}$ such that, for all $(x, y^+, y^-) \in \mathcal{D}$,*

$$\log \pi_\theta(y^+ | x) - \log \pi_\theta(y^- | x) \approx f(\psi(y^+)) - f(\psi(y^-)), \quad (24)$$

with small approximation error, while the latent geometry remains unseparated:

$$\cos(p, c) \approx \cos(p, r). \quad (25)$$

Interpretation. This definition isolates a policy whose output ordering follows a surface feature while its prompt-response similarity margin remains small. The output margin alone does not reveal that representational condition. DSLA applies pressure to the similarity margin directly, providing an additional training constraint. A shortcut can itself be encoded geometrically, so satisfying the latent margin is not sufficient to identify the intended preference feature.

Lemma 2 (Bounded latent margins). *For all nonzero $p_i, c_i, r_i, d_i, \bar{d}_B$, the similarity margin and directional margin satisfy*

$$-2 \leq s_i(\theta) \leq 2, \quad -1 \leq a_i(\theta) \leq 1. \quad (26)$$

Proof. Cosine similarity lies in $[-1, 1]$ by Cauchy–Schwarz, so the difference of two cosine terms lies in $[-2, 2]$. The directional margin is itself a cosine similarity between two normalized vectors, and hence also lies in $[-1, 1]$. $\square$

Interpretation. The latent margins have a fixed geometric range, which bounds the corresponding losses for fixed m and γ . Their gradients depend on the margin Jacobians and on γ ; a bounded margin does not imply a uniform bound on the parameter gradient. The normalization makes angular relationships the principal object of supervision.

Lemma 3 (Approximate scale invariance away from the stabilizer). *In the idealized limit $\varepsilon \rightarrow 0$, the latent losses $\mathcal{L}_{\text{sim}}$ and $\mathcal{L}_{\text{dir}}$ are invariant to isotropic rescaling of the hidden states: for any $\alpha > 0$,*

$$\mathcal{L}_{\text{sim}}(\alpha H) = \mathcal{L}_{\text{sim}}(H), \quad \mathcal{L}_{\text{dir}}(\alpha H) = \mathcal{L}_{\text{dir}}(H). \quad (27)$$

With the implementation’s positive numerical stabilizers, this equality is approximate whenever the relevant vector norms dominate the stabilizers. The losses are not invariant under general non-orthogonal linear transformations.

Proof. When $\varepsilon = 0$, cosine similarity depends only on the angle between vectors, so $\cos(\alpha a, \alpha b) = \cos(a, b)$ for all $\alpha > 0$. The batch direction u_B in Eq. 21 is likewise scale-invariant. The implementation uses positive stabilizers inside the square-root normalizations and in the batch-direction denominator; these perturb the equality only when a relevant norm is comparable to the corresponding stabilizer. A non-orthogonal linear map changes angles, so invariance fails in general. $\square$

Interpretation. When the vector norms dominate the stabilizers, rescaling the hidden states leaves the latent objective approximately unchanged. The optimizer must therefore alter angular relationships to increase the margins. Near zero, the stabilizers keep the objective defined and introduce a dependence on vector magnitude.

Lemma 4 (Soft-margin upper bound). *For any scalar margin z and target margin m , the loss $\text{softplus}(\gamma(m - z))$ is a convex, differentiable upper bound on the hinge loss $\max\{0, \gamma(m - z)\}$ up to an additive $\log 2$:*

$$\max\{0, \gamma(m - z)\} \leq \text{softplus}(\gamma(m - z)) \leq \max\{0, \gamma(m - z)\} + \log 2. \quad (28)$$

Proof. For any u , $\max\{0, u\} \leq \log(1 + \exp u) \leq \max\{0, u\} + \log 2$. Substituting $u = \gamma(m - z)$ gives the claim. $\square$

Interpretation. The softplus loss supplies a differentiable relaxation of the geometric margin constraint. The bound quantifies its relation to the hinge penalty, including the additive approximation gap at the margin boundary.

Proposition 1 (Latent pressure survives saturated preference probabilities). *Assume the model parameters decompose locally into a hidden-state map ϕ_θ and a readout W , and suppose a training example satisfies $\beta\Delta_{\text{DPO},i}(\theta) \gg 0$ so that the DPO gradient contribution is exponentially small. If $s_i(\theta) < m$ or $a_i(\theta) < 0$, then for $\lambda > 0$ and $\gamma > 0$ the DSLA gradient contains a nonzero latent component whenever the corresponding margin has nonzero Jacobian with respect to θ .*

Proof. The DPO derivative is proportional to $-\beta\sigma(-\beta\Delta_{\text{DPO},i})$, which tends to zero exponentially as the preference margin grows. By contrast,

$$\frac{\partial}{\partial s_i} \text{softplus}(\gamma(m - s_i)) = -\gamma\sigma(\gamma(m - s_i)), \quad (29)$$

which is strictly negative for finite s_i . Similarly,

$$\frac{\partial}{\partial a_i} \text{softplus}(-\gamma a_i) = -\gamma\sigma(-\gamma a_i), \quad (30)$$

which is strictly negative for finite a_i . Therefore, if the latent margin has a nonzero parameter Jacobian, the chain rule yields a nonzero gradient contribution scaled by λ . $\square$

Interpretation. The proposition identifies when latent supervision supplies pressure that remains after the DPO output margin has saturated. A correctly ranked response can still violate the similarity constraint or disagree with the batch direction. The latent term continues to act on that violation whenever its parameter Jacobian is nonzero.

Proposition 2 (No expected gradient cross-term under orthogonality). *Let $g_{\text{pref}} = \nabla_\theta \mathcal{L}_{\text{pref}}$ and $g_{\text{lat}} = \nabla_\theta \mathcal{L}_{\text{latent}}$. If $\mathbb{E}[\langle g_{\text{pref}}, g_{\text{lat}} \rangle] = 0$, then the expected squared norm of the combined gradient contains no cross-term:*

$$\mathbb{E}[\|g_{\text{pref}} + \lambda g_{\text{lat}}\|_2^2] = \mathbb{E}[\|g_{\text{pref}}\|_2^2] + \lambda^2 \mathbb{E}[\|g_{\text{lat}}\|_2^2]. \quad (31)$$

Proof. Expand the squared norm and take expectations. The cross-term is $2\lambda\mathbb{E}[\langle g_{\text{pref}}, g_{\text{lat}} \rangle]$, which vanishes by assumption. $\square$

Interpretation. The orthogonality assumption removes the expected cross-term in the squared update norm. This identity characterizes the combined gradient under that assumption; it does not establish that the assumption holds during training. The empirical comparisons examine whether latent supervision is compatible with the output objective in the reported settings.

Theorem 1 (Conditional directional recovery). *Suppose there exists a unit vector $u_\star \in \mathbb{R}^d$ such that the pairwise latent differences satisfy*

$$d_i = \mu u_\star + \xi_i, \quad \mathbb{E}[\xi_i] = 0, \quad \|\xi_i\|_2 \leq R, \quad (32)$$

for $\mu > 0$, and let $u_B = \bar{d}_B / \|\bar{d}_B\|_2$ be the batch direction. If $|B| = b$ and $\|\bar{\xi}_B\|_2 \leq \mu/2$, then

$$\langle u_B, u_\star \rangle \geq 1 - \frac{2\|\bar{\xi}_B\|_2^2}{\mu^2}. \quad (33)$$

Moreover, under bounded independent noise, $\|\bar{\xi}_B\|_2 = O(R/\sqrt{b})$ with high probability, so the estimated direction improves at the usual mini-batch averaging rate.

Proof. Write $\bar{d}_B = \mu u_\star + \bar{\xi}_B$. The angle between $\bar{d}_B$ and $u_\star$ is controlled by the component of $\bar{\xi}_B$ orthogonal to $u_\star$. Since $\|\mu u_\star + \bar{\xi}_B\|_2 \geq \mu - \|\bar{\xi}_B\|_2 \geq \mu/2$, a standard perturbation bound for normalized vectors gives

$$\|u_B - u_\star\|_2 \leq \frac{2\|\bar{\xi}_B\|_2}{\mu}.$$

Using $\|u_B - u_\star\|_2^2 = 2(1 - \langle u_B, u_\star \rangle)$ yields the claim. The high-probability rate follows from vector-valued concentration for bounded independent noise. $\square$

Interpretation. Under the shared-direction model, averaging reduces the contribution of independent pair noise and improves the directional estimate. When the displacements have no stable shared component, the theorem’s premise does not hold and increasing batch size alone need not recover a useful direction. This distinction motivates examining batch composition together with batch size.

Corollary 1 (Degeneracy at batch size one). *When $b = 1$ and $d_i \neq 0$, the directional margin satisfies $a_i \approx 1$ up to numerical stabilization. Consequently, $\mathcal{L}_{\text{dir}}$ carries little discriminative information at batch size one.*

Proof. For a singleton batch, $u_B = d_i / \|d_i\|_2$. Substituting into Eq. 22 gives the inner product of a normalized vector with itself, equal to one up to the ε used for numerical stability. $\square$

Interpretation. Many reported runs use batch size one, where the direction term contributes little beyond stabilization. Improvements in those runs establish an effect of the combined training objective, but cannot establish the benefit of agreement across multiple preference pairs. Ordinary gradient accumulation preserves this degeneracy if the latent objective is computed separately in each singleton microbatch. To estimate a shared direction across accumulated examples, their representations must be combined before computing u_B and $\mathcal{L}_{\text{dir}}$.

Proposition 3 (Conditional readout-level improvement). *Let the final logits be $z(h) = Wh + b$ for a readout $W \in \mathbb{R}^{V \times d}$. Suppose there exists a unit vector $u_\star \in \mathbb{R}^d$, a convex set $\mathcal{C} \subseteq \mathbb{R}^d$, and a constant $\kappa > 0$ such that, for every hidden state $h \in \mathcal{C}$, the directional derivative of the preferred-versus-rejected logit margin $\Delta_{\text{logit}}(h)$ along $u_\star$ satisfies*

$$\langle \nabla \Delta_{\text{logit}}(h), u_\star \rangle \geq \kappa. \quad (34)$$

If a DSLA update keeps all pooled hidden states in $\mathcal{C}$ and increases $\mathbb{E}[\langle c_i - r_i, u_\star \rangle]$ by $\delta > 0$ without changing W or b , then the expected logit margin increases by at least $\kappa\delta$.

Proof. Because $\mathcal{C}$ is convex, the line segment between the old and updated hidden states of each example lies in $\mathcal{C}$. By the mean-value theorem and the stated directional derivative bound, the pointwise increase in Δ_{logit} is at least κ times the increase in $\langle c_i - r_i, u_\star \rangle$. Taking expectations yields the bound. $\square$

Interpretation. The result connects latent displacement to output behavior when the assumed direction exists and the local readout uses it. The semantic meaning of that direction depends on the dataset and model. Establishing that it represents a particular property, such as truthfulness or helpfulness, requires evidence beyond the geometric condition.

6 Ablation Studies

We evaluate how explicit latent supervision changes preference training. Matched comparisons hold the model, dataset, example budget, batch size, and validation conditions fixed while varying the objective. We use output losses to characterize optimization, hidden-state diagnostics to measure the geometric effect, and generation diagnostics to examine its consequences. The layer, weight, and pooling ablations then vary the representation and strength of the additional supervision. Section 7 states the scope of the evaluation.

The experiments use Qwen2.5 [18], Qwen3 [19], and Llama 3 [20], paired with Human-Like-DPO [21] and JOSIE-2-Instruct-DPO. The first corpus provides a compact setting for studying preference-margin formation. The second is a 1K hand-curated JOSIEfication corpus whose chosen responses encode the creative, direct, and honest assistant style used in our previous Josiefied-Qwen3 models. The datasets let us compare latent supervision under different preference criteria, including an explicitly specified assistant persona.

We compare DPO with DSLA-DPO and ORPO with DSLA-ORPO. Smaller and mid-size backbones are evaluated on both datasets; larger 4-bit backbones use smaller example budgets, as identified in the comparison figures. The main protocol uses 128 held-out validation examples, 16 generation samples, validation every 100 training steps, and seed = 42. The controlled-shortcut study uses the separate protocol specified in Section 6.9.

6.1 Experimental Setup

Models. We fine-tune instruction-tuned Qwen2.5, Qwen3, and Llama 3 checkpoints, including full-precision and 4-bit variants. The comparison matrix covers differences in model family, parameter scale, and quantization. Training uses the MLX backend [22].

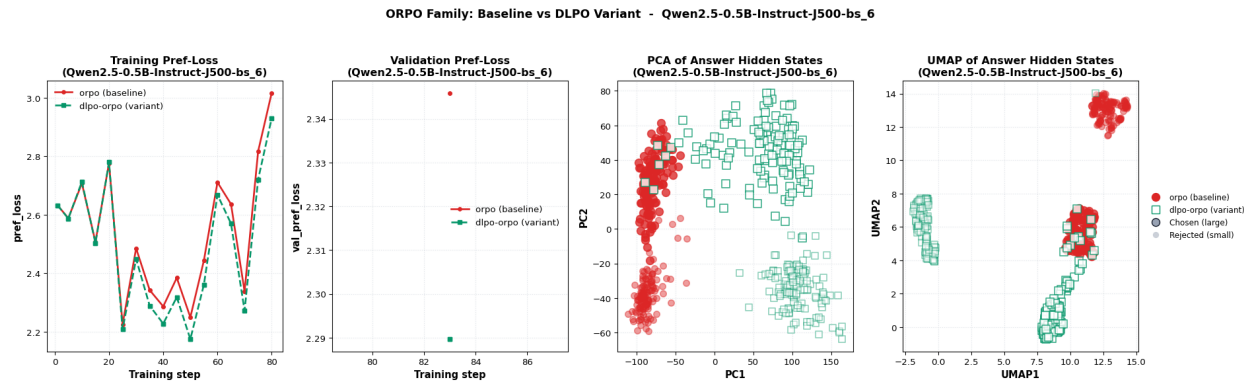


Figure 2: Matched ORPO and DSLA-ORPO training on the JOSIE preference setting at batch size six. DSLA-ORPO reaches lower training and held-out validation loss than the ORPO baseline while producing a distinct answer-state geometry. The figure supports a within-batch-size comparison between objectives; it does not by itself identify a causal effect of batch size.

Datasets. We evaluate Human-Like-DPO and JOSIE-2-Instruct-DPO at a common 500-example budget in the principal small-model comparisons. The JOSIE corpus contains 1K custom preference pairs designed to express the target assistant persona. Matching the example budget controls training exposure while allowing the preference criterion to vary. The main protocol holds out 128 examples, evaluates every 100 steps over 16 validation batches, and uses seed = 42.

Algorithms and step budgets. We train two objective families: the DPO family, consisting of the probability-only dpo baseline and its latent-regularized DSLA-DPO extension, and the ORPO family, consisting of the orpo baseline and its DSLA-ORPO extension. The baseline runs set the latent regularizer to $\lambda = 0.0$ (the DSLA code path with the regularizer disabled), while the DSLA variants activate the prompt-response and preference-direction terms described above. We specify step budgets in examples rather than raw optimizer steps, and convert each example budget into optimizer steps by dividing by batch size; this keeps the number of training examples approximately fixed across batch-size ablations.

Shared hyperparameters. Across the runs we use a learning rate of 1×10^{-5} , DPO $\beta = 0.1$, ORPO $\alpha = 0.1$, maximum sequence length 2048, LoRA rank 8, LoRA scale 20.0, LoRA dropout 0.0, and LoRA applied to 16 transformer layers. The latent objective uses `latent_margin = 0.05`, `latent_gamma = 10.0`, `latent_variant = both`, answer-mean pooling, and final-layer hidden states. We save checkpoints every 50 steps, log training metrics every 5 steps, and log validation metrics every 100 steps.

6.2 Loss Dynamics

We track validation loss, preference accuracy, policy margin, and latent similarity margin together. Small preference datasets can reach high accuracy while leaving substantial differences in the hidden-state geometry. In the DPO runs, output-margin saturation makes the latent diagnostics particularly useful: they show whether the representation continues to change after the output loss becomes small. In the ORPO runs, we examine how latent supervision changes chosen–rejected separation alongside the combined supervised and preference objective.

Matched batch-6 comparison. Figure 2 compares ORPO and DSLA-ORPO on Qwen2.5-0.5B with the JOSIE dataset at batch size six. With the backbone, example budget, seed, and optimizer configuration matched, DSLA-ORPO finishes with lower training and validation loss and higher held-out preference accuracy. A batch containing multiple pairs allows the directional term to compare their displacements. The comparison is consistent with a useful contribution from the latent objective, although attributing the result specifically to the direction term requires a component ablation.

6.3 Quantitative Comparisons

Tables 1 and 2 report the generation diagnostics for the completed adapters, grouped by objective family. Each row uses 16 validation-generation examples from the adapter’s training dataset. The column labeled F1 Fraction is the

Model	Dataset	Algorithm	F1 Fraction	F1 Gap	LogP Gap	Rep Rate	Gen Len	Latent Dist
Qwen2.5-0.5B-Instruct	Human-Like-DPO	DSLA-DPO	0.9375	0.0861	226.1532	0.0000	127.9	0.6931
Qwen2.5-0.5B-Instruct	Human-Like-DPO	dpo	0.6250	0.0709	293.8988	0.0312	122.9	0.1042
Qwen2.5-0.5B-Instruct	JOSIE-2-DPO	DSLA-DPO	0.0000	-0.0574	-2244.4810	0.0018	112.1	0.4655
Qwen2.5-0.5B-Instruct	JOSIE-2-DPO	dpo	0.0625	-0.0587	-3094.9810	0.0000	103.2	0.0850

Table 1: DPO-family generation diagnostics for DPO and DSLA-DPO adapters. Each row uses 16 validation examples from its training dataset; comparisons pair adapters within the same model and dataset setting. F1 Fraction counts generations with greater token F1 against the chosen reference than the rejected reference. It is a lexical agreement metric. Gap columns report chosen-side minus rejected-side scores. Bold marks the larger F1 fraction, generation length, or latent distance within a pair; greater length or distance alone does not establish better response quality.

Model	Dataset	Algorithm	F1 Fraction	F1 Gap	LogP Gap	Rep Rate	Gen Len	Latent Dist
Llama-3.1-8B-Instruct-4bit	Human-Like-DPO	DSLA-ORPO	1.0000	0.1454	136.0977	0.0013	124.2	0.4154
Llama-3.1-8B-Instruct-4bit	Human-Like-DPO	orpo	0.9375	0.1390	159.4949	0.0007	123.9	0.2980
Llama-3.1-8B-Instruct-4bit	JOSIE-2-DPO	DSLA-ORPO	0.0625	-0.0375	-1960.8396	0.0008	121.6	0.1570
Llama-3.1-8B-Instruct-4bit	JOSIE-2-DPO	orpo	0.0000	-0.0606	-2037.7446	0.0046	116.8	0.0663
Llama-3.2-1B-Instruct-4bit	Human-Like-DPO	DSLA-ORPO	1.0000	0.1885	265.0759	0.0013	122.5	0.2957
Llama-3.2-1B-Instruct-4bit	Human-Like-DPO	orpo	1.0000	0.1723	313.9608	0.0006	125.0	0.2289
Qwen2.5-0.5B-Instruct	Human-Like-DPO	DSLA-ORPO	0.9375	0.1321	621.2128	0.0006	128.0	0.5287
Qwen2.5-0.5B-Instruct	Human-Like-DPO	orpo	1.0000	0.1485	463.9534	0.0006	128.0	0.1914
Qwen2.5-0.5B-Instruct	JOSIE-2-DPO	DSLA-ORPO	0.0000	-0.0563	11282.7199	0.0026	125.2	0.8789
Qwen2.5-0.5B-Instruct	JOSIE-2-DPO	orpo	0.0000	-0.0382	-2362.8359	0.0274	127.8	0.0496
Qwen2.5-1.5B-Instruct	Human-Like-DPO	DSLA-ORPO	1.0000	0.1590	510.3052	0.0000	128.0	0.3710
Qwen2.5-1.5B-Instruct	Human-Like-DPO	orpo	1.0000	0.1495	357.7912	0.0000	127.8	0.1104
Qwen2.5-1.5B-Instruct	JOSIE-2-DPO	DSLA-ORPO	0.1250	-0.0287	12658.9735	0.0128	128.1	0.8912
Qwen2.5-1.5B-Instruct	JOSIE-2-DPO	orpo	0.0625	-0.0496	7299.5049	0.0008	114.4	0.2714
Qwen2.5-3B-Instruct	Human-Like-DPO	DSLA-ORPO	1.0000	0.1771	276.7966	0.0000	126.2	0.1528
Qwen2.5-3B-Instruct	Human-Like-DPO	orpo	1.0000	0.1806	312.7577	0.0000	127.2	0.0630
Qwen2.5-7B-Instruct-4bit	Human-Like-DPO	DSLA-ORPO	1.0000	0.1833	144.1209	0.0000	118.2	0.2186
Qwen2.5-7B-Instruct-4bit	Human-Like-DPO	orpo	1.0000	0.1910	188.3907	0.0006	117.7	0.1269
Qwen3-4B-Instruct-2507-4bit	Human-Like-DPO	DSLA-ORPO	1.0000	0.1531	114.7762	0.0000	120.4	0.2036
Qwen3-4B-Instruct-2507-4bit	Human-Like-DPO	orpo	1.0000	0.1525	109.6570	0.0006	118.7	0.0820
Qwen3-4B-Instruct-2507-4bit	JOSIE-2-DPO	DSLA-ORPO	0.0625	-0.0485	-2014.8568	0.0063	123.2	0.0579
Qwen3-4B-Instruct-2507-4bit	JOSIE-2-DPO	orpo	0.1250	-0.0444	-2018.8466	0.0054	124.1	0.0413

Table 2: ORPO-family generation diagnostics for ORPO and DSLA-ORPO adapters. Each row uses 16 validation examples from its training dataset; comparisons pair adapters within the same model and dataset setting. F1 Fraction counts generations with greater token F1 against the chosen reference than the rejected reference. It is a lexical agreement metric. Gap columns report chosen-side minus rejected-side scores. Bold marks the larger F1 fraction, generation length, or latent distance within a pair; greater length or distance alone does not establish better response quality.

fraction of generated answers with greater token F1 against the chosen reference than against the rejected reference. It measures lexical reference agreement; response quality requires a separate judgment. The remaining columns report reference-score gaps, repetition, generation length, and chosen-rejected latent distance.

Every reported matched comparison has a larger latent distance under DSLA. In the DPO family, the distance increases from 0.1042 to 0.6931 on Human-Like-DPO and from 0.0850 to 0.4655 on JOSIE-2-DPO. DSLA-ORPO increases distance in all eleven ORPO comparisons. On the JOSIE corpus, Qwen2.5-0.5B increases from 0.0496 to 0.8789, and Qwen2.5-1.5B from 0.2714 to 0.8912. These measurements establish the effect of the latent objective on chosen-rejected separation under the reported extraction protocol.

The generated-reference overlap diagnostic varies across settings. DSLA-DPO increases its chosen-reference F1 fraction on Human-Like-DPO (0.6250 $\rightarrow$ 0.9375) and decreases it on JOSIE (0.0625 $\rightarrow$ 0.0000). DSLA-ORPO increases the fraction in three comparisons, ties ORPO in six, and decreases it in two. The consistent geometric effect therefore accompanies mixed generation outcomes. Section 6.10 examines this difference in more detail.

6.4 Hidden-State Geometry

Table 3 isolates the subset of model-dataset pairs for which all four algorithms are available. This stricter comparison removes a large amount of heterogeneity: the backbone, dataset, validation-evaluation size, and evaluation protocol are fixed, and only the preference objective family and latent regularization status vary. Under this restriction, the largest latent distance always belongs to a DSLA method. On Human-Like-DPO, the largest distance is produced by

Model	Dataset	DPO Dist	DSLA-DPO Dist	ORPO Dist	DSLA-ORPO Dist
Qwen2.5-0.5B-Instruct	Human-Like-DPO	0.1042	0.6931	0.1914	0.5287
Qwen2.5-0.5B-Instruct	JOSIE-2-DPO	0.0850	0.4655	0.0496	0.8789

Table 3: Four-algorithm latent-distance comparison for model–dataset pairs where DPO, DSLA-DPO, ORPO, and DSLA-ORPO are all available. Distances are the averaged Latent Dist values from the same 16-example validation-evaluation summaries used in Tables 1 and 2. Bold values mark the largest latent distance within each model–dataset group.

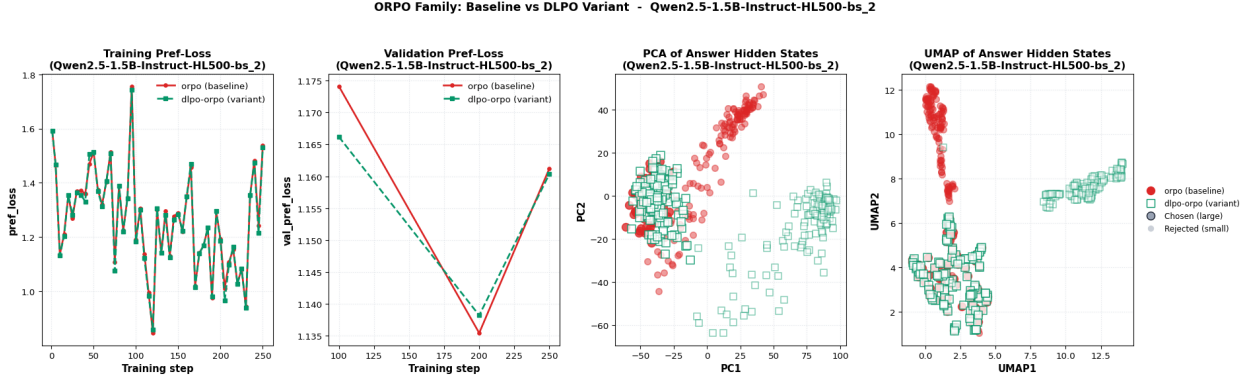


Figure 3: ORPO versus DSLA-ORPO on Qwen2.5-1.5B-Instruct trained on the Human-Like-DPO 500-example setting. Preference-loss curves remain close, while PCA and UMAP projections show a materially different latent organization under DSLA-ORPO.

DSLA-DPO (0.6931), exceeding both plain DPO (0.1042) and plain ORPO (0.1914), while also exceeding DSLA-ORPO (0.5287). On JOSIE-2-DPO, the largest distance is produced by DSLA-ORPO (0.8789), exceeding DSLA-DPO (0.4655), DPO (0.0850), and ORPO (0.0496).

In the complete comparison subset, a DSLA variant produces the largest latent distance on each dataset. The objective family associated with that maximum changes: DSLA-DPO gives the greatest separation on Human-Like-DPO, while DSLA-ORPO gives the greatest separation on JOSIE. One possible explanation is that ORPO’s supervised component interacts differently with the persona criterion than DPO’s reference-relative objective. The comparison motivates a controlled study of that interaction; it does not identify its cause.

The Latent Dist metric summarizes chosen–rejected separation under the extraction and pooling protocol. Greater distance establishes a change in that geometry, but does not identify the quality or behavioral meaning of the separated features. We therefore use it alongside the output and shortcut diagnostics. The measurements in this section use the final residual stream, with extraction-depth sensitivity examined in Section 6.5.

Figures 3, 4, and 5 illustrate representative ORPO comparisons. On Qwen2.5-1.5B with Human-Like-DPO, the loss curves remain close while the PCA and UMAP layouts differ. On JOSIE, the same backbone shows a stronger displacement between the ORPO and DSLA-ORPO representations. The Llama-3.2-1B projections remain more interleaved, consistent with its smaller distance increase from 0.2289 to 0.2957. These figures show that the visible geometric effect varies with the model and dataset. Projection layouts alone do not establish which architecture has greater capacity for behavioral control.

6.5 Layer-Anchor Ablation

We vary the extraction layer to determine how the depth receiving latent supervision affects training and representation geometry. In the Qwen2.5-0.5B Human-Like-DPO 600-example setting, we compare middle, late, and final anchors for DSLA-ORPO and DSLA-DPO while holding the remaining configuration fixed. Figure 6 shows that the late and final anchors provide more favorable optimization or projected separation than the middle anchor. Extraction depth is therefore a substantive choice in this setting.

Observed depth sensitivity. The late and final anchors offer different advantages. The final anchor produces stable loss curves and clear projected separation, while the late anchor reaches the lowest ORPO validation loss. The middle

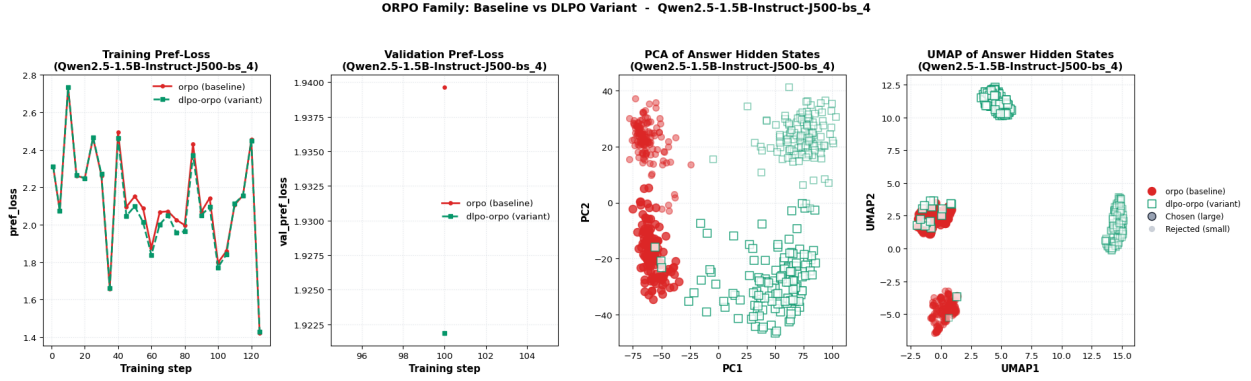


Figure 4: ORPO versus DSLA-ORPO on Qwen2.5-1.5B-Instruct trained on the J0SIE-2-DPO 500-example setting. The representation projections show stronger separation than the corresponding loss curves, suggesting that the latent term changes the internal organization of the style-bearing preference.

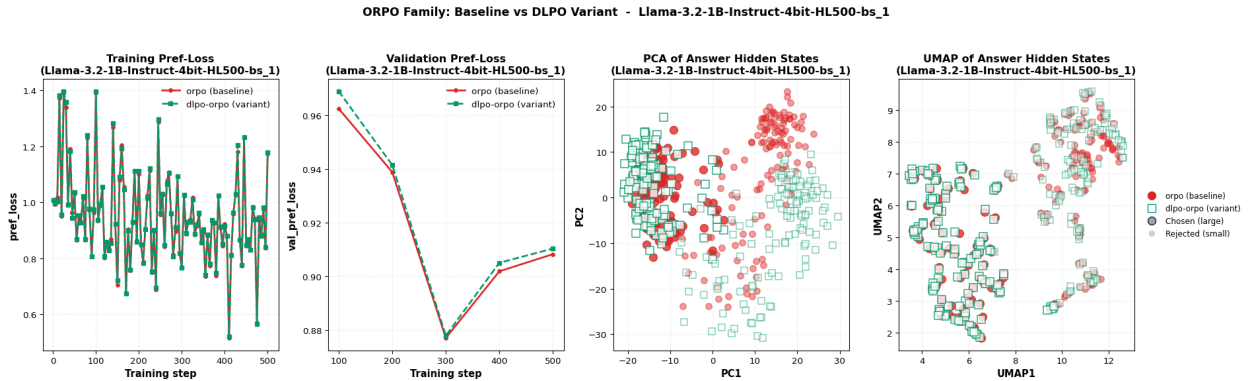


Figure 5: ORPO versus DSLA-ORPO on Llama-3.2-1B-Instruct-4bit in the Human-Like-DPO 500-example setting. Loss curves remain closely matched, while the latent projections show less separation than the Qwen2.5-1.5B comparisons.

anchor has a noisier early trajectory and less distinct projected geometry. This pattern suggests that later representations are better suited to the tested latent objective. Because the experiment changes where supervision is applied, it identifies sensitivity to intervention depth rather than proving that preference information exists only at those layers.

ORPO-family depth sensitivity. For DSLA-ORPO, the middle anchor starts near 4.8 training preference loss and falls to roughly 2.2 over the reported trajectory. The final anchor trains more smoothly and reaches validation loss near 1.5. The late anchor reaches the lowest validation loss, approximately 0.06 at the final checkpoint. The PCA and UMAP panels also depend on extraction depth: the middle configuration is more entangled, while the final configuration shows clearer separation. Loss and projected geometry thus favor different aspects of the late and final configurations.

DPO-family depth sensitivity. For DSLA-DPO, validation preference loss approaches zero across all three anchors after the early checkpoints. The middle anchor shows larger early training spikes, but the final scalar losses provide little distinction among the configurations. Their projected representations still differ: rejected states occupy depth-dependent regions, and the final configuration shows clearer chosen–rejected separation in UMAP. Saturation of the output objective therefore leaves extraction-depth effects visible in the representation diagnostics.

Mechanistic interpretation. Behavioral-subspace methods motivate examining depth because their interventions depend on the representations available at the selected layer [7, 6]. Our ablation extends that question to latent preference supervision. A possible explanation for the observed depth sensitivity is that later residual states expose features more

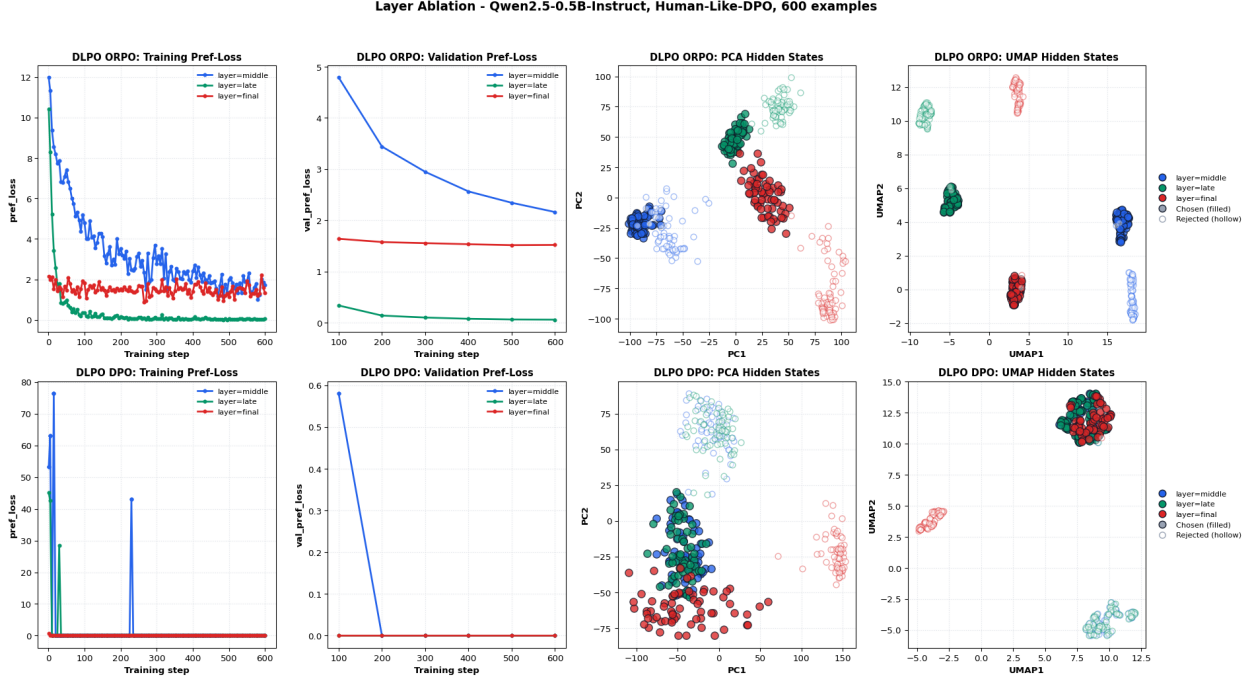


Figure 6: Extraction-layer ablation on Qwen2.5-0.5B-Instruct in the Human-Like-DPO 600-example setting. Each row compares middle, late, and final anchors within a DSLA objective family. The final anchor shows stable optimization and clear projected separation; the late anchor reaches the lowest ORPO validation loss. The comparison establishes sensitivity to the depth receiving latent supervision in this setting.

directly useful to the output decision. The present experiment does not trace when those features form or establish that the same depth ordering holds across behaviors and architectures.

Practical conclusion. We retain the final residual stream as the default for the broader comparison matrix. In this ablation, the late anchor is a strong alternative for ORPO when selecting by validation loss, while the final anchor gives clearer projected separation for DPO. The middle anchor is less favorable under the tested configuration. We recommend selecting the extraction layer with both output and representation diagnostics; the ordering reported here applies to this model and dataset. The subsequent weight and pooling studies use the final layer for comparability.

6.6 Latent-Weight Ablation

The latent-weight ablation isolates the regularization coefficient λ in Eq. 14. We sweep $\lambda \in \{0, 0.05, 0.1, 0.2, 0.4\}$ in the same Qwen2.5-0.5B Human-Like-DPO 600-example setting, keeping the dataset, seed, LoRA configuration, batch size, layer anchor, and validation protocol fixed. Figure 7 reports the resulting train-loss, validation-loss, PCA, and UMAP diagnostics for both DSLA-ORPO and DSLA-DPO. This ablation is particularly important because it separates two questions that are otherwise easy to conflate: whether increasing λ improves the probability-level objective, and whether increasing λ changes the hidden geometry even when the probability-level objective barely moves.

For DSLA-ORPO, the five training preference-loss curves nearly overlap, and validation losses decrease from roughly 1.64 to the 1.51–1.52 range. The small differences favoring $\lambda = 0.2$ or $\lambda = 0.4$ do not establish a robust tuning preference. The projected representations are more sensitive: rejected states occupy weight-dependent regions, while the $\lambda = 0.4$ UMAP layout shows broader chosen-state dispersion and several outliers. The coefficient changes the geometry even when the output loss changes little.

For DSLA-DPO, training preference loss approaches zero after the initial transient at every weight, and validation loss is plotted at the 10^{-5} scale. The PCA and UMAP layouts nevertheless remain weight-dependent, especially in the organization of rejected states. The scalar objective has insufficient resolution to characterize those differences once its margin saturates.

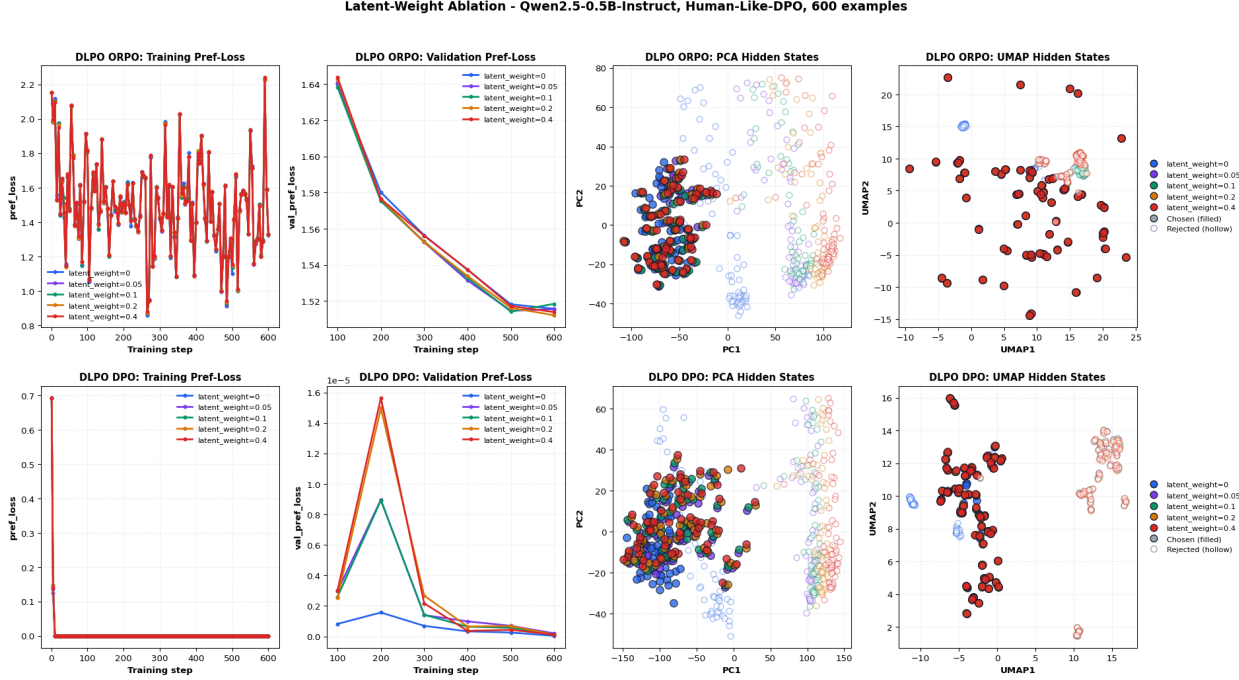


Figure 7: Latent-weight ablation on Qwen2.5-0.5B-Instruct trained on Human-Like-DPO for 600 examples, sweeping $\lambda \in \{0, 0.05, 0.1, 0.2, 0.4\}$. The ORPO-family losses are nearly invariant to λ while the PCA and UMAP geometries change substantially, and the DPO-family losses saturate near zero even though hidden-state organization remains weight-dependent. The figure therefore supports the central methodological point that DSLA must be evaluated through both scalar preference losses and representation-space diagnostics.

We use $\lambda = 0.1$ as a working default in this setting: it produces a visible geometric effect without the broader projected dispersion observed at $\lambda = 0.4$. The sweep supports evaluating regularization strength with representation diagnostics alongside validation loss. Establishing whether the larger dispersion is beneficial or harmful requires behavioral evaluation across additional runs.

6.7 Pooling-Operator Ablation

Pooling constructs the response vector from the token-level hidden states. It determines whether the latent objective reads the entire answer, the final active token, a terminal window, or the combined prompt and answer. We compare these choices on Qwen2.5-0.5B-Instruct in the Human-Like-DPO 600-example setting, using

$$\mathcal{P} \in \{\text{answer_mean}, \text{last_token}, \text{last_k_mean}, \text{prompt_answer_mean}\} \quad (35)$$

for DSLA-ORPO and DSLA-DPO. We hold the dataset, seed, LoRA configuration, batch size, extraction layer, and $\lambda = 0.1$ fixed. Figure 8 shows the loss curves and projections; Table 4 reports the final-checkpoint metrics.

For DSLA-ORPO, validation losses remain close across poolers: 1.520, 1.516, 1.521, and 1.519 for `answer_mean`, `last_token`, `last_k_mean`, and `prompt_answer_mean`, respectively. Last-token pooling reaches preference accuracy 0.9375 and similarity margin 0.886, compared with accuracy 0.875 and margins in the 0.51–0.54 range for the alternatives. Its projected rejected states also overlap less with the chosen states. These observations favor last-token pooling in this particular comparison. Concentrating direct supervision on the terminal position is a possible explanation for the result.

For DSLA-DPO, all poolers reach 1.0 validation preference accuracy and losses in the 10^{-4} – 10^{-5} range. Their latent similarity margins are also close, as Table 4 shows, while the projections differ in the locations and clustering of rejected states. Last-token pooling shows clearer UMAP separation, and prompt-plus-answer pooling shows more interleaving. Including the prompt in the response pool changes the quantity compared with the prompt vector, so its margin should be interpreted under that altered representation definition.

Algorithm	Pooler	val_pref_acc	val_loss	val_sim_margin
DSLA-ORPO	answer_mean	0.875	1.5200	0.513
DSLA-ORPO	last_token	0.9375	1.5155	0.886
DSLA-ORPO	last_k_mean	0.875	1.5210	0.538
DSLA-ORPO	prompt_answer_mean	0.875	1.5187	0.540
DSLA-DPO	answer_mean	1.000	$3.46 \cdot 10^{-5}$	0.275
DSLA-DPO	last_token	1.000	$3.27 \cdot 10^{-4}$	0.272
DSLA-DPO	last_k_mean	1.000	$4.95 \cdot 10^{-4}$	0.273
DSLA-DPO	prompt_answer_mean	1.000	$2.54 \cdot 10^{-5}$	0.275

Table 4: Final-checkpoint pooling metrics on Qwen2.5-0.5B-Instruct in the Human-Like-DPO 600-example setting. Last-token pooling gives the highest DSLA-ORPO preference accuracy and similarity margin. DSLA-DPO reaches saturated output loss and similar latent margins across the four poolers.

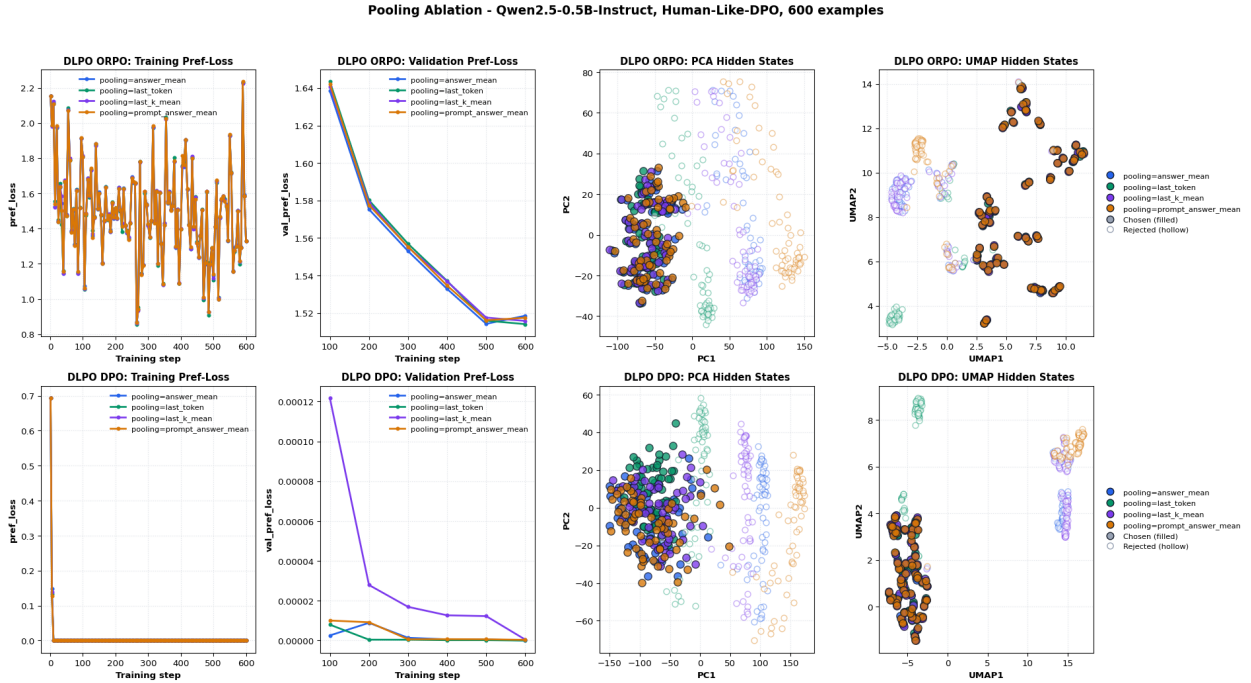


Figure 8: Pooling-operator ablation on Qwen2.5-0.5B-Instruct trained on Human-Like-DPO for 600 examples, sweeping the pooler over $\{ \text{answer_mean}, \text{last_token}, \text{last_k_mean}, \text{prompt_answer_mean} \}$ for both DSLA-ORPO and DSLA-DPO. The ORPO-family runs are nearly indistinguishable on scalar loss but resolve cleanly on the latent similarity margin and on the geometry panels, with `last_token` pooling producing the largest final-checkpoint margin and the cleanest UMAP cluster structure. The DPO-family runs saturate the probability objective regardless of pooler, yet the answer-state geometry remains visibly pooler-dependent.

Pooling determines where the latent loss applies its direct gradient. With last-token pooling, the correction enters through the final active token; with answer-mean pooling, Lemma 1 distributes it across all active response positions. This difference explains how the pooler changes the intervention, although it does not by itself explain the measured performance ordering. We retain answer-mean pooling as the default because the constraint then supervises the pooled response span. Last-token pooling is a competitive alternative in the reported ORPO setting. Its behavior on longer responses remains to be evaluated.

The pooling study establishes sensitivity to how a response representation is constructed. Last-token pooling gives the highest ORPO preference accuracy and similarity margin in this setting, while DPO’s saturated output loss offers little basis for choosing among poolers. Comparing the choices across models, datasets, and response lengths would establish the scope of the observed advantage.

6.8 Shortcut-Feature Geometry

Increased chosen–rejected separation leaves open which features account for that separation. We examine this question by measuring the association between response-level surface features and the preference-direction projection. The observational analysis uses the held-out 128-example split of Human-Like-DPO; the controlled study in Section 6.9 introduces specific textual perturbations. Correlations in the original hidden-state space provide the main diagnostic, while UMAP coordinates offer a view of the projected layout.

Setup. For each cell we recompute the answer hidden states from both the orpo and the DSLA-ORPO adapter on the held-out evaluation split used by the orpo-vs-DSLA-ORPO figure. We then define, for each adapter, the mean preference direction

$$u_{\text{pref}}(\theta) = \frac{1}{|\mathcal{V}|} \sum_{i \in \mathcal{V}} [H_{\theta}(x_i, y_i^+) - H_{\theta}(x_i, y_i^-)], \quad (36)$$

where $\mathcal{V}$ indexes the validation pairs and each vector is an answer-pooled hidden state from the selected extraction layer. For each response, $p_i = \langle H_i, u_{\text{pref}} \rangle$ gives its projection onto this estimate. We compute Pearson correlations between each surface feature and the first UMAP coordinate, the second UMAP coordinate, and $|p_i|$. The last quantity measures the feature’s linear association with projection magnitude; a small correlation does not establish independence or identify what the direction encodes.

Shortcut feature definitions. We compute four scalar features from response text, excluding the prompt. *Length* counts response tokens after appending the end-of-sequence token. *Markdown density* counts formatting markers—code fences, headings, bold markers, bullets, and blockquotes—and divides by token count. *Hedge density* counts case-insensitive occurrences of maybe, perhaps, possibly, likely, might, may, I think, I believe, I guess, and I suppose, divided by token count. *List density* is the fraction of non-empty lines beginning with a dash, asterisk, or numbered list marker. These transparent heuristics summarize candidate surface proxies; they do not measure response quality or the complete set of possible shortcuts.

Dataset-level shortcut deltas. The Human-Like-DPO corpus carries a strong length bias: on the held-out 128-example evaluation split, the chosen responses average 193.77 tokens and the rejected responses average 135.82 tokens, a chosen-vs-rejected delta of +57.95 tokens. Markdown density is small in absolute terms and slightly higher on the rejected side (−0.0008 per token). Hedge density is mildly higher on the chosen side (+0.0022 per token). The largest dataset-level shortcut signal is list density: the rejected responses are roughly an order of magnitude more list-heavy than the chosen responses (rejected 0.0497, chosen 0.0051, a chosen-vs-rejected delta of −0.0446). These four deltas are intrinsic to the dataset; they are identical whether measured under the orpo or the DSLA-ORPO adapter, because both adapters see the same input texts. The question the correlations answer is therefore not “does the dataset contain a shortcut gradient” (it does) but “does the algorithm’s learned preference direction in hidden space align with that gradient”.

Quantitative reading of the correlation table. Under ORPO, length has the largest absolute correlation with $|p_i|$ on both backbones: +0.563 for Qwen2.5-1.5B and +0.424 for Qwen2.5-0.5B. DSLA-ORPO reduces these values to +0.312 and +0.235. Hedge correlations also decrease, from +0.265 to +0.208 and from +0.229 to +0.195. The remaining features do not follow a uniform reduction: markdown correlations remain small, while the magnitude of the list correlation increases under DSLA on both backbones. The result therefore concerns selected measured features, especially length. Correlations with individual UMAP coordinates depend on the embedding layout and are not invariant measures of alignment in the original hidden space.

Variation across model scales. The length-correlation reduction is larger for the 1.5B model (+0.563 → +0.312, a change of −0.251) than for the 0.5B model (+0.424 → +0.235, a change of −0.189). Hedge correlation follows the same ordering, with reductions of 0.057 and 0.034. These observations motivate studying model scale as a factor in the geometric response to latent supervision. The pair of backbones and single training seed do not isolate capacity as the cause of the difference.

Backbone	Algorithm	Feature	$\rho(\text{UMAP}_1)$	$\rho(\text{UMAP}_2)$	$\rho(p_i)$
Qwen2.5-1.5B-Instruct	orpo	length	+0.058	−0.504	+0.563
	orpo	markdown	+0.044	−0.039	+0.063
	orpo	hedge	+0.126	−0.225	+0.265
	orpo	list	+0.063	−0.047	+0.041
	DSLA-ORPO	length	−0.504	−0.381	+0.312
	DSLA-ORPO	markdown	−0.064	−0.006	−0.059
	DSLA-ORPO	hedge	−0.231	−0.165	+0.208
	DSLA-ORPO	list	+0.023	+0.098	−0.188
Qwen2.5-0.5B-Instruct	orpo	length	+0.312	−0.413	+0.424
	orpo	markdown	+0.108	−0.030	−0.013
	orpo	hedge	+0.204	−0.202	+0.229
	orpo	list	+0.130	−0.029	−0.134
	DSLA-ORPO	length	−0.432	+0.456	+0.235
	DSLA-ORPO	markdown	−0.012	+0.018	−0.088
	DSLA-ORPO	hedge	−0.225	+0.242	+0.195
	DSLA-ORPO	list	+0.101	−0.073	−0.234

Table 5: Pearson correlations between surface features and UMAP coordinates or absolute hidden-state projection $|p_i|$, using the direction estimate in Eq. 36. UMAP is fitted to the union of the four response groups. DSLA-ORPO lowers the length correlation from +0.563 to +0.312 on Qwen2.5-1.5B and from +0.424 to +0.235 on Qwen2.5-0.5B. Other features have different patterns, including increased absolute list correlation.

6.9 Controlled Shortcut Study

We complement the observational correlations with controlled preference pairs constructed through specific surface edits. Each corpus varies a selected textual feature between otherwise closely matched responses. This design makes the perturbation explicit and reduces differences in response content, allowing us to examine its association with the learned projection. A surface edit can still change tokenization and correlated features, so the study measures a response to the constructed perturbation rather than establishing the removal of all confounding effects.

Constructed datasets. We construct three preference corpora, each with 91 pairs, through small edits to existing responses. Each edit introduces a selected surface difference between the chosen and rejected roles. The responses remain close in average length, with the measured deltas reported below.

- **markdown.** Chosen responses use ****bold**** markdown markers; rejected responses are identical apart from stripping those markers. Engineered shortcut: *markdown density*.
- **format.** Chosen responses include extra list-item markers (`\n-`); rejected responses are otherwise identical. Engineered shortcut: *list density*.
- **word.** Chosen responses open with a Hey! greeting prefix; rejected responses are identical apart from that prefix. Engineered shortcut: *length* (the prefix adds ~ 2 tokens).

On the held-out 16-example split, the selected features have the following chosen-minus-rejected deltas: markdown density +0.0481 per token (chosen 0.0583, rejected 0.0102), list density +0.3999 (chosen 0.7138, rejected 0.3139), and length +2.13 tokens (chosen 257.38, rejected 255.25). The remaining measured features are approximately balanced between response roles; this does not eliminate correlations among feature values across examples.

Setup. For each of the three datasets we fine-tuned Qwen2.5-0.5B-Instruct under both orpo and DSLA-ORPO with identical hyperparameters to the rest of the ablation: LoRA rank 8, scale 20.0, applied to 16 layers; batch size 2; 30 iters over 80 training pairs and 16 validation pairs; learning rate 1×10^{-5} ; $\beta = 0.1$, $\alpha = 0.1$, $\lambda = 0.1$, $m = 0.05$, $\gamma = 10.0$; final-layer answer-mean pooling; seed = 42. After training we recomputed the answer hidden states of both adapters on the 16-example validation split, fitted a single UMAP on the union of all four groups per cell, and measured the per-feature Pearson correlation between the four shortcut features and three target variables: the two UMAP coordinates and the absolute projection $|\langle H_i, u_{\text{pref}} \rangle|$ of the response’s hidden state onto the algorithm-specific mean preference direction of Eq. 36. This protocol is identical to §6.8 apart from the source of the evaluation pairs.

Final-step validation metrics. The two algorithms reach comparable probability-level performance on every dataset, confirming that the DSLA-ORPO latent regularizer does not impede probability-level learning at this scale: markdown

Dataset (engineered shortcut)	Algorithm	Feature	$\rho(\text{UMAP}_1)$	$\rho(\text{UMAP}_2)$	$\rho(p_i)$
markdown (markdown density)	orpo	length	+0.328	+0.250	−0.519
	orpo	markdown	−0.335	−0.295	+0.190
	orpo	hedge	+0.185	+0.137	−0.299
	orpo	list	+0.438	+0.273	−0.570
	DSLA-ORPO	length	+0.328	+0.234	−0.438
	DSLA-ORPO	markdown	−0.345	−0.294	+0.166
	DSLA-ORPO	hedge	+0.179	+0.134	−0.233
	DSLA-ORPO	list	+0.444	+0.237	−0.499
format (list density)	orpo	length	−0.699	−0.406	−0.540
	orpo	markdown	−0.316	+0.071	−0.306
	orpo	hedge	−0.108	−0.180	−0.168
	orpo	list	−0.495	−0.369	−0.165
	DSLA-ORPO	length	−0.673	−0.403	−0.516
	DSLA-ORPO	markdown	−0.292	+0.070	−0.315
	DSLA-ORPO	hedge	−0.088	−0.182	−0.218
	DSLA-ORPO	list	−0.508	−0.361	−0.108
word (length)	orpo	length	−0.422	+0.592	−0.514
	orpo	markdown	+0.092	+0.165	−0.329
	orpo	hedge	−0.150	−0.046	+0.016
	orpo	list	−0.571	+0.778	−0.533
	DSLA-ORPO	length	−0.401	+0.589	−0.474
	DSLA-ORPO	markdown	+0.097	+0.211	−0.346
	DSLA-ORPO	hedge	−0.128	−0.088	−0.053
	DSLA-ORPO	list	−0.547	+0.759	−0.490

Table 6: Surface-feature correlations on each controlled corpus’s 16-example validation split, using the protocol of Table 5. Bold marks the feature targeted by the edit. Its absolute correlation with $|p_i|$ decreases under DSLA-ORPO in each comparison: markdown $0.190 \rightarrow 0.166$, list formatting $0.165 \rightarrow 0.108$, and greeting-associated length $0.514 \rightarrow 0.474$. These are geometric point estimates under the constructed perturbations.

val_pref_acc is 0.625 under both; format improves from 0.4375 under orpo to 0.625 under DSLA-ORPO; word is 0.9375 under both. The DSLA-ORPO latent terms are active on all three cells: final-step validation $\rho(|p_i|)$ for the engineered shortcut is below its counterpart under the orpo baseline in every cell (see Table 6 below).

Reading. The absolute correlation between the selected feature and $|p_i|$ decreases in each controlled comparison:

- On the markdown cell the engineered shortcut is *markdown density*. Under orpo the absolute projection correlation is +0.190; under DSLA-ORPO it is +0.166, a reduction of 0.024 (about 13%) without sign change. Length drops from −0.519 to −0.438.
- On the format cell the engineered shortcut is *list density*. Under orpo the correlation is −0.165; under DSLA-ORPO it is −0.108, a reduction of 0.057 (about 35%) without sign change. Length drops from −0.540 to −0.516.
- On the word cell the engineered shortcut is *length*. Under orpo the correlation is −0.514; under DSLA-ORPO it is −0.474, a reduction of 0.040 (about 8%) without sign change.

The selected feature’s absolute correlation decreases in all three controlled comparisons, matching the direction of the length-correlation reduction in the natural corpus. The controlled reductions are smaller. Model scale, the 30-step training budget, and the 80 training examples are possible contributors, but the study does not isolate their effects. The result is a repeated pattern in the reported point estimates, with residual feature correlations remaining in each configuration.

Summary. Across the observational and controlled comparisons, DSLA-ORPO lowers the absolute correlation between $|p_i|$ and the selected surface feature. The pattern supports a change in how that feature relates to the measured preference geometry. It does not establish that the model has replaced the feature with a semantic quality criterion or that its generations are less susceptible to shortcut exploitation. Those questions require behavioral evaluations that hold response content and the intended criterion under independent control.

6.10 Generative Evaluation

Generation length changes alongside the latent intervention. DSLA-DPO produces longer outputs in both matched comparisons; DSLA-ORPO is longer in six comparisons, tied in one, and shorter in four. We treat length as a diagnostic because its usefulness depends on the task and response content. The measurements show a shift in generation behavior, but additional length alone does not establish improved quality.

The chosen-reference F1 fraction and F1 gap depend on the dataset and backbone. On Human-Like-DPO, several ORPO comparisons already reach 1.0000 on the overlap fraction and leave no headroom for that metric. On JOSIE, the fractions are much lower: DSLA-ORPO increases them for Llama-3.1-8B and Qwen2.5-1.5B, ties Qwen2.5-0.5B, and decreases them for Qwen3-4B. These are differences in lexical reference agreement under a small generation sample. Direct judgments of persona fidelity, helpfulness, or reasoning would be needed to evaluate the corresponding response properties.

The generation diagnostics and latent distances reveal different aspects of the training effect. chosen-rejected separation increases in every reported matched comparison, while generated-reference agreement has gains, ties, and regressions. The objective therefore provides a repeatable geometric intervention in this comparison matrix whose generation consequences depend on the setting.

7 Limitations

The experiments establish the behavior of DSLA under the reported training and evaluation conditions. The following limitations determine how far those observations can be generalized.

Single random seed. The main comparison matrix uses $\text{seed} = 42$ and the latent configuration $(\lambda, m, \gamma) = (0.1, 0.05, 10.0)$. The component ablations vary their specified factors while retaining the seed. Tables 1–2 therefore report single-seed point estimates without standard errors. Replication across training seeds is needed to estimate the variability of the measured effects.

Held-out evaluation size. Each generation row uses a 16-example sample. This supports inspection of the reported matched outcomes, while sampling uncertainty limits conclusions about the wider prompt distribution. The F1 Fraction column measures chosen-reference token-F1 agreement and cannot substitute for a judged preference evaluation.

Hyperparameter robustness. We do not sweep the margin m or sharpness γ ; $m = 0.05$ and $\gamma = 10.0$ are working defaults. The weight study in Section 6.6 varies $\lambda \in \{0, 0.05, 0.1, 0.2, 0.4\}$ on a Qwen2.5-0.5B-Instruct Human-Like-DPO 600-example setting. Its findings characterize that configuration. Joint layer, weight, and pooling studies across additional datasets are needed to establish broader robustness.

Benchmark-level evidence. We do not evaluate DSLA on a benchmark suite (e.g., MT-Bench, AlpacaEval, MMLU, or a held-out reasoning benchmark). The current evidence is geometric and matched-generation, not benchmark-level. Establishing that the latent-term’s representational effect translates into a benchmark-level preference win requires a separate evaluation campaign.

Dataset size and diversity. The two datasets used in this study (Human-Like-DPO and JOSIE-2-Instruct-DPO) are small (at most 1K pairs) and reflect a narrow distribution of preferences. The transfer of the geometric findings to larger, more diverse preference corpora—in particular, corpora that span code, math, and long-context reasoning—is an open question. The three controlled-shortcut corpora of §6.9 are even smaller (91 pairs each); they are designed as a controlled test of the shortcut-geometry claim, not as a benchmark for transfer.

Shortcut-feature proxy. The surface features are heuristic proxies, and their correlations with $|p_i|$ describe an association with projection magnitude. Reduced absolute correlation does not establish independence, removal of a shortcut, or a more semantic representation. The controlled edits can also change multiple textual measurements despite targeting a selected feature. Evaluations with independent quality labels and controlled feature changes are needed to connect these geometric observations to behavioral shortcut dependence.

Causal intervention tests. We do not perform causal intervention tests on the learned preference direction. Establishing that the DSLA-induced direction is causally implicated in preference behavior would require activation-steering

or ablation probes at inference time, analogous to those used in the Gabliteration and Abliteration lines of work. The present paper establishes the existence of a learned direction; the causal role of that direction is a separate question.

Interpreting preference geometry. The latent objective inherits the preference information available in the dataset and representation. Truthfulness, safety, helpfulness, and reasoning quality cannot be inferred from directional agreement alone. A heterogeneous corpus can also contain preference criteria that are poorly represented by a shared batch direction. The mathematical results state the relevant assumptions, and the experiments show the measured geometric and generation effects under the evaluated conditions.

8 Conclusion

We introduced Directional and Similarity-aware Latent Alignment to make hidden-state relationships an explicit part of preference optimization. The similarity margin supervises the chosen response relative to its prompt and rejected counterpart, while the directional term encourages agreement among within-batch preference displacements. The formulation integrates with DPO and ORPO, and its analysis characterizes the geometric margins, their approximate scale invariance, and the latent pressure that remains after output-margin saturation.

The reported matched comparisons show increased chosen–rejected separation under DSLA, including settings with closely matched output-loss curves. The shortcut studies show reduced correlation with selected surface features, and the layer, weight, and pooling ablations identify choices that change the resulting geometry. Generation diagnostics vary across settings, leaving the relationship between latent separation and response quality as a central empirical question.

Further work should examine training-seed variability, larger and more diverse preference corpora, and generation quality under independent evaluation. Activation-steering and direction-ablation experiments can test whether the learned geometry participates causally in preference behavior. Joint extraction and regularization studies can then determine when a shared direction is useful and when the objective should represent multiple criteria. These questions follow directly from the intervention DSLA makes available: explicit control over how preference supervision acts on the model’s hidden representations.

Appendix Overview

This appendix collects the formal matched-comparison convention used throughout the experimental section (§A), additional representation-space diagnostics including centered displacement and directional coherence (§B), dataset and training details (§C), reproducibility invariants (§D), and supplementary ORPO-family visual comparisons (§E).

A Extended Experimental Results

We report ablation results only under matched experimental conditions. For any scalar metric m computed from a run, we define a matched comparison tuple

$$\tau = (\mathcal{M}, \mathcal{D}, \mathcal{A}, B, T, s), \quad (37)$$

where $\mathcal{M}$ is the base model, $\mathcal{D}$ is the dataset, $\mathcal{A} \in \{\text{DPO}, \text{ORPO}\}$ is the output-level objective family, B is the batch size, T is the example budget, and s is the random seed. The effect of the DSLA regularizer on metric m is therefore not defined as an unconditional difference across all runs, but as the paired contrast

$$\Delta_m(\tau) = m(\text{DSLA-}\mathcal{A}; \tau) - m(\mathcal{A}; \tau). \quad (38)$$

This convention prevents comparisons in which a change in backbone, dataset, batch size, or training horizon is incorrectly attributed to the latent regularizer.

When multiple matched tuples are available, we summarize the direction of the effect with the empirical mean

$$\bar{\Delta}_m = \frac{1}{|\mathcal{T}|} \sum_{\tau \in \mathcal{T}} \Delta_m(\tau), \quad (39)$$

and retain the tuple-level values rather than reducing the entire study to a single aggregate score. This is necessary because DSLA is explicitly a representation-level intervention; a positive average effect can hide dataset-specific failures, while a neutral average can hide systematic improvements in one objective family and degradation in another.

B Additional Representation Analysis

Let h_i^+ and h_i^- denote the pooled hidden states for the chosen and rejected responses of example i , and let

$$d_i = h_i^+ - h_i^- \quad (40)$$

be the example-level preference displacement. The batch preference direction used by the latent objective is

$$\bar{d}_B = \frac{1}{B} \sum_{i=1}^B d_i, \quad u_B = \frac{\bar{d}_B}{\|\bar{d}_B\|_2 + \epsilon}, \quad (41)$$

where $\epsilon > 0$ prevents numerical instability when the batch displacement is close to zero. A natural diagnostic for directional coherence is the mean signed projection

$$S_B = \frac{1}{B} \sum_{i=1}^B \langle d_i, u_B \rangle, \quad (42)$$

which separates two phenomena that are otherwise conflated: the magnitude of chosen–rejected separation and the extent to which examples agree on a shared direction.

We also consider a centered variant that removes a batch-level common mode:

$$\tilde{h}_i^+ = h_i^+ - \mu_B, \quad \tilde{h}_i^- = h_i^- - \mu_B, \quad \mu_B = \frac{1}{2B} \sum_{i=1}^B (h_i^+ + h_i^-). \quad (43)$$

The centered displacement $\tilde{d}_i = \tilde{h}_i^+ - \tilde{h}_i^-$ is algebraically equal to d_i , but the centered states are useful when visualizing or probing whether chosen and rejected responses separate because of a genuine pairwise displacement or because all completions drift together under fine-tuning.

If the displacements decompose as

$$d_i = \alpha_i u_\star + \xi_i, \quad \mathbb{E}[\xi_i] = 0, \quad \mathbb{E}[\langle \xi_i, u_\star \rangle] = 0, \quad (44)$$

then the batch mean in Eq. 41 estimates the shared direction u_* up to sampling noise. Under bounded noise $\|\xi_i\|_2 \leq \sigma$, the deviation of $\bar{d}_B$ from its expectation scales with batch size as

$$\mathbb{E} \left[\|\bar{d}_B - \mathbb{E}[d_i]\|_2^2 \right] = \frac{1}{B} \mathbb{E} \left[\|d_i - \mathbb{E}[d_i]\|_2^2 \right], \quad (45)$$

which gives the informal justification for treating batch size as a substantive ablation variable rather than merely as an optimization convenience.

C Dataset and Training Details

Each preference dataset is represented as

$$\mathcal{D} = \{(x_i, y_i^+, y_i^-)\}_{i=1}^n, \quad (46)$$

where x_i is the instruction or prompt, y_i^+ is the preferred completion, and y_i^- is the dispreferred completion. For a total example budget T and batch size B , the number of optimizer iterations is

$$N_{\text{steps}} = \max \left(1, \left\lceil \frac{T}{B} \right\rceil \right), \quad (47)$$

so that batch-size ablations approximately preserve the number of training examples seen by the optimizer. This convention is essential for interpreting batch-size effects: otherwise, larger batches would silently receive a larger example budget or smaller batches would receive more parameter updates.

The validation split is denoted

$$\mathcal{D}_{\text{val}} \subset \mathcal{D}, \quad |\mathcal{D}_{\text{val}}| = 128 \quad (48)$$

for the current ablation protocol. For any validation metric m , we write

$$\hat{m}_{\text{val}} = \frac{1}{|\mathcal{D}_{\text{val}}|} \sum_{(x_i, y_i^+, y_i^-) \in \mathcal{D}_{\text{val}}} m(x_i, y_i^+, y_i^-), \quad (49)$$

and interpret it as a finite-sample diagnostic rather than a population estimate of alignment quality.

D Reproducibility Notes

We identify each run by the tuple

$$\rho = (\text{model_id}, \text{dataset_repo}, \text{dataset_file}, \text{algorithm}, T, B, s, \lambda), \quad (50)$$

where $\lambda = 0$ for probability-only baselines and $\lambda > 0$ for DSLA variants. A result is considered comparable only when all fields except the algorithm and latent weight are matched according to the contrast in Eq. 38. In particular, a DSLA run should not be compared against a baseline run trained with a different dataset revision, validation split, batch size, seed, tokenizer, maximum sequence length, or LoRA configuration.

The minimal reproducibility invariant for a reported metric is therefore

$$\text{Comparable}(\rho_1, \rho_2) = \mathbf{1}[\rho_1 \setminus \{\text{algorithm}, \lambda\} = \rho_2 \setminus \{\text{algorithm}, \lambda\}], \quad (51)$$

where the notation removes the fields varied between a baseline and its DSLA extension. Matching the remaining fields controls the recorded training conditions when attributing a difference to the latent objective.

E Additional ORPO-Family Visual Comparisons

We provide four additional ORPO and DSLA-ORPO comparisons from the experimental matrix. Each figure shows training preference loss, validation preference loss, PCA, and UMAP of answer hidden states. Together, they illustrate how closely matched output-loss trajectories can accompany different projected representations across backbones.

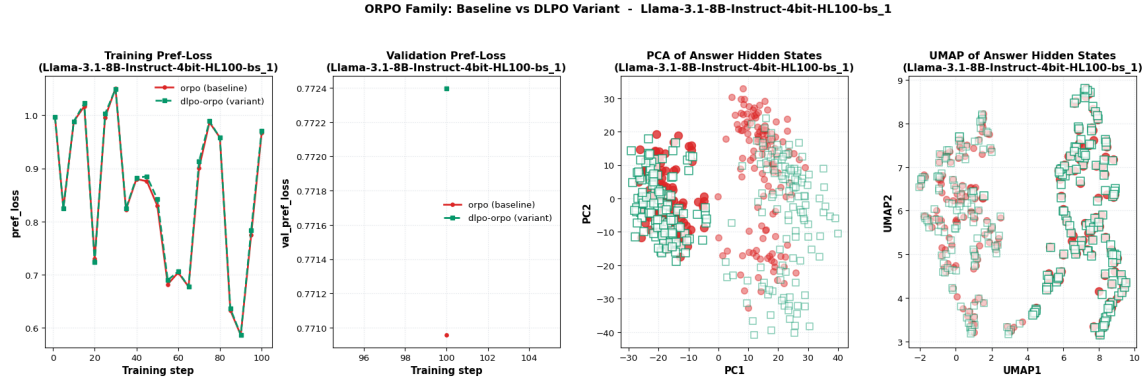


Figure 9: Additional visual comparison for Llama-3.1-8B-Instruct-4bit on Human-Like-DPO. The training curves remain close, while the projection panels show moderate DSLA-ORPO displacement relative to ORPO.

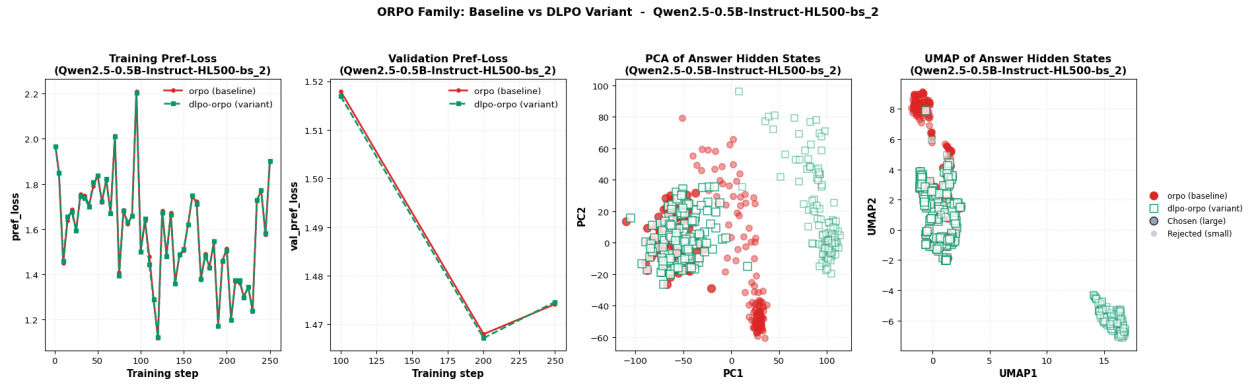


Figure 10: Additional visual comparison for Qwen2.5-0.5B-Instruct on Human-Like-DPO. The DSLA-ORPO projection separates from ORPO despite similar optimization traces, matching the larger latent distance reported in Table 2.

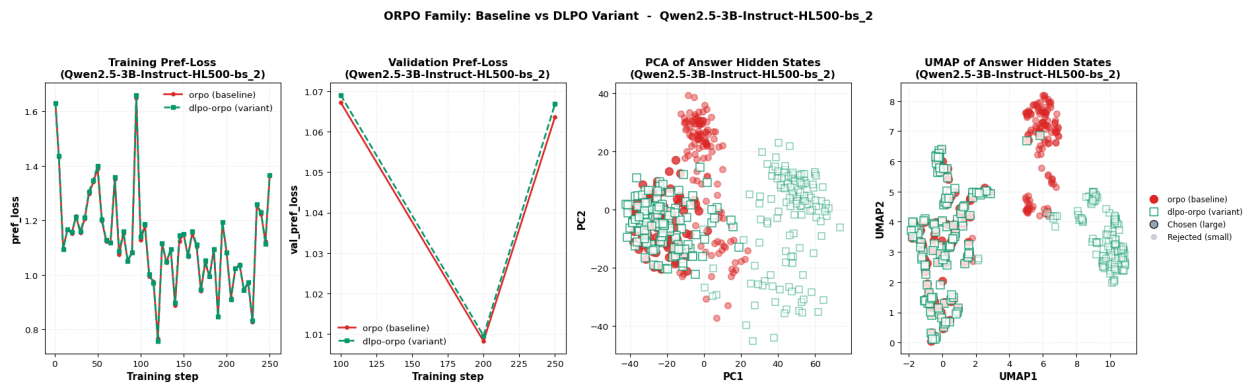


Figure 11: Additional visual comparison for Qwen2.5-3B-Instruct on Human-Like-DPO. The projection panels show DSLA-induced movement in answer-state geometry, while the loss panels remain baseline-like, reinforcing the separation between scalar objective behavior and representational organization.

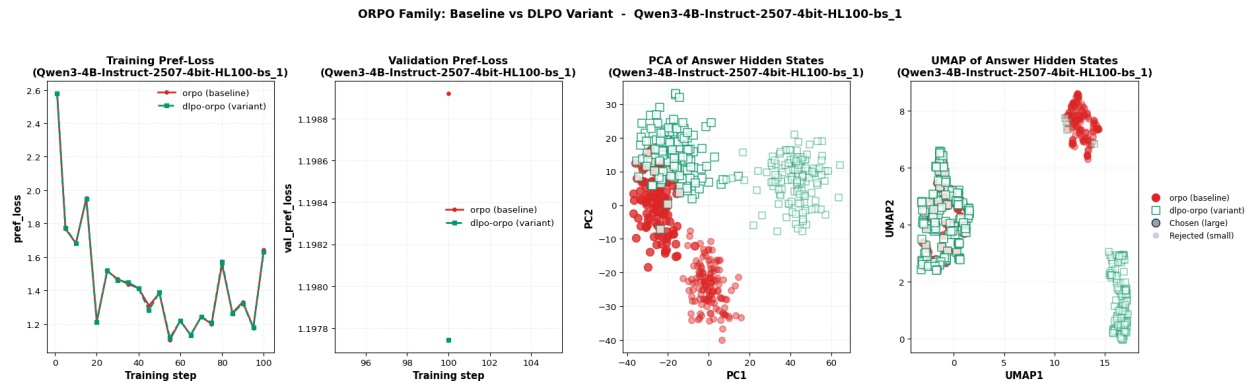


Figure 12: Additional visual comparison for Qwen3-4B-Instruct-2507-4bit on Human-Like-DPO. DSLA-ORPO changes the answer-state layout while leaving high-level loss behavior close to ORPO, consistent with the paper’s main representational claim.

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